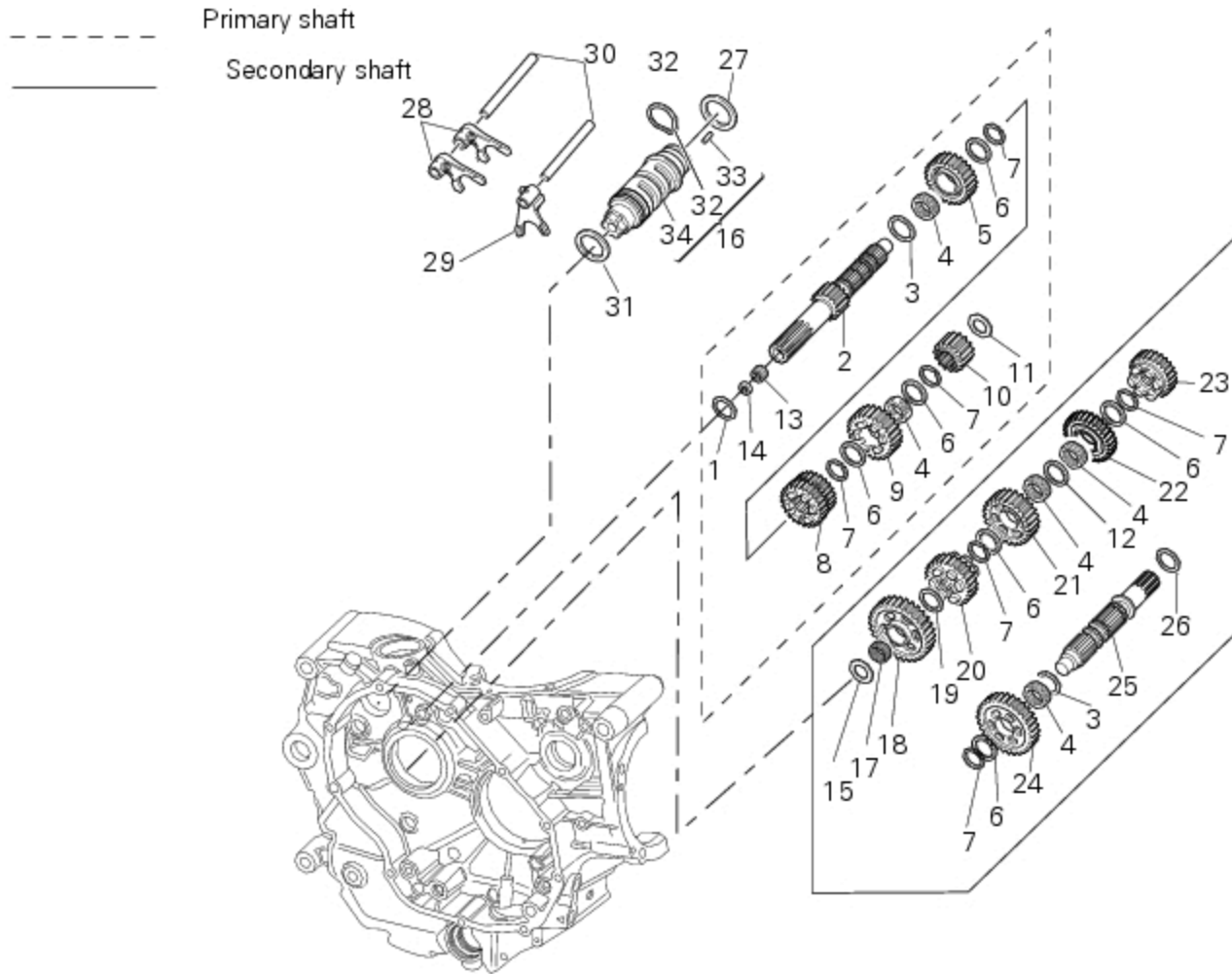


7.2 -Gearbox assembly: gearbox shafts



- 1 Shim, thickness 1
- 2 Gearbox primary shaft
- 3 Shim, thickness 0.5
- 4 Needle roller bearing
- 5 5th speed driving gear
- 6 Splined washer, thickness 0.5
- 7 Circlip
- 8 3rd- 4th speed driving gear
- 9 6th speed driving gear
- 10 2nd speed driving gear
- 11 Shim, thickness 1.8
- 12 Splined washer, thickness 0.5
- 13 Roller bearing
- 14 Sealing ring
- 15 Shim washer
- 16 Selector drum assembly
- 17 Needle roller bearing
- 18 1st speed driven gear
- 19 Shim, thickness 1
- 20 5th speed driven gear
- 21 4th speed driven gear
- 22 3rd speed driven gear
- 23 6th speed driven gear
- 24 2nd speed driven gear

- 25 Gearbox secondary shaft
- 26 Shim washer
- 27 Shim, thickness 1
- 28 1st, 4th, 2nd, 3rd speed selector fork
- 29 5th, 6th speed selector fork
- 30 Selector fork shaft
- 31 Shim, thickness 1
- 32 Needle roller retaining circlip (square cross-section)
- 33 Special needle roller (square cross-section)
- 34 Selector drum

 Spare parts catalogue

- 1100 ABS [GEARCHANGE CONTROL](#)
- 1100 ABS [GEARBOX](#)
- 1100 S ABS [GEARCHANGE CONTROL](#)
- 1100 S ABS [GEARBOX](#)



Important

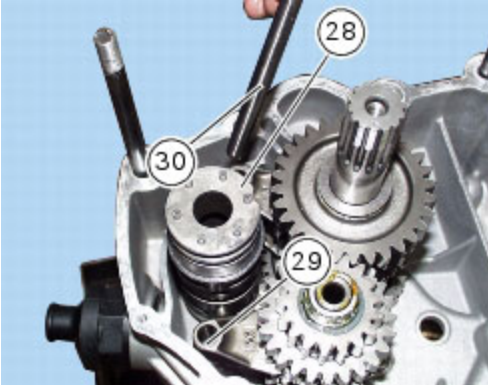
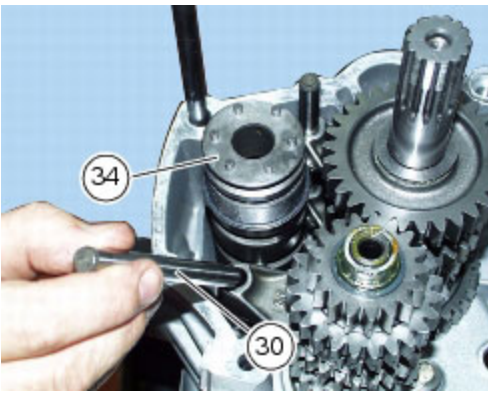
Bold reference numbers in this section identify parts not shown in the figures alongside the text, but which can be found in the exploded view diagram.

Removal of the gearbox assembly

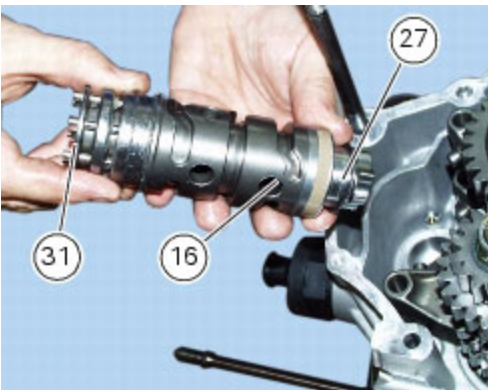
Operation	Section reference
Remove the engine from the frame	N 1, Removal of the engine
Remove the lubrication system	N 2.1, Removal of the oil pump
Remove the timing belts and the timing belt pulleys	N 4.2, Removal of the timing belt covers
Remove the camshafts	N 4.3, Removal of the camshafts
Remove the complete cylinder head unit	N 4.4, Removal of the cylinder head assembly
Remove the cylinder barrel/piston assemblies	N 5, Removal of the cylinder/piston assembly
Remove the alternator-side crankcase cover and the alternator assembly	N 8, Removal of the generator cover
Remove the mesh filter	D 4, Changing the engine oil and filter cartridge
Remove the complete clutch assembly	N 6.1, Removal of the clutch
Remove the clutch-side crankcase cover	N 6.2, Removal of the clutch cover
Remove the primary gear	N 6.3, Removal of the primary drive gear
Remove the starter motor cam chain idler gear	N 9.1, Removal of the starter motor idler gear
Remove the starter motor	P 3, Removal of the starter motor
Separate the crankcase halves	N 9.2, Separation of the crankcase halves

Withdraw the selector fork shafts (30).

Move the forks (28) and (29) to disengage them from the slots in the selector drum (34).



Withdraw the selector drum (16) taking care not to lose shims (31) and (27) mounted on the shaft. Note that the positions of the shims must not be inverted.

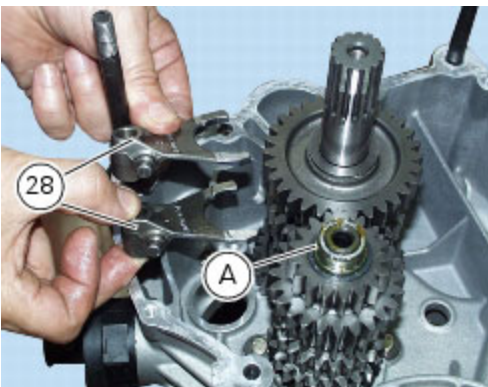


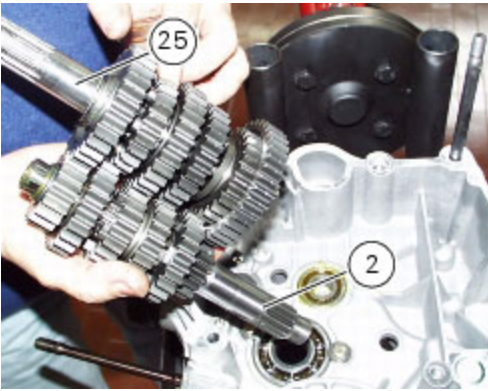
Now you can renew the needle roller retaining ring (**32**) and the special needle rollers (**33**).

Remove gear selector forks (**29**) and (28).

Remove the gearbox primary (2) and secondary (25) shafts complete with gears, taking care to recover the spacers on the ends of the shafts.

If the bearing inner rings (A) are left on the shafts, slide them off the ends of the gearbox primary (2) and secondary (25) shafts (Sect. N 9.2, [Overhauling the main bearings](#)).





Disassembly of the gearbox shafts

Place the shaft in a vice in such a way as to facilitate the disassembly operations.

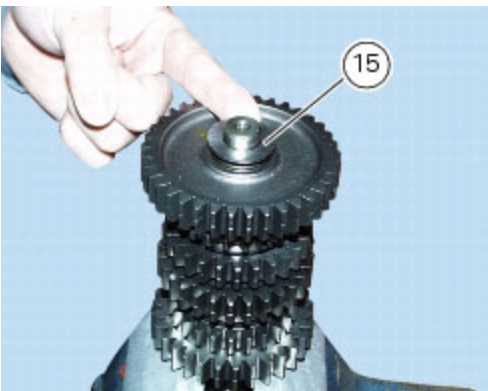


Important

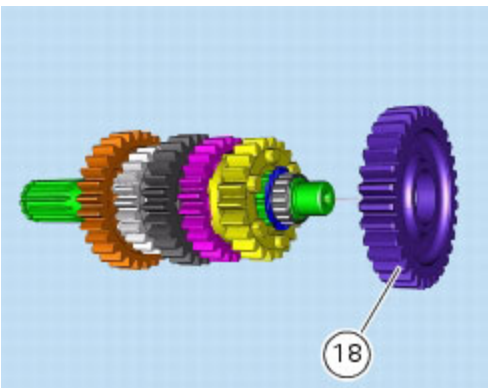
Take care not to invert the positions of the shims on reassembly: this would potentially lead to jamming when using the gear selector control, making it necessary to reopen the engine crankcase.

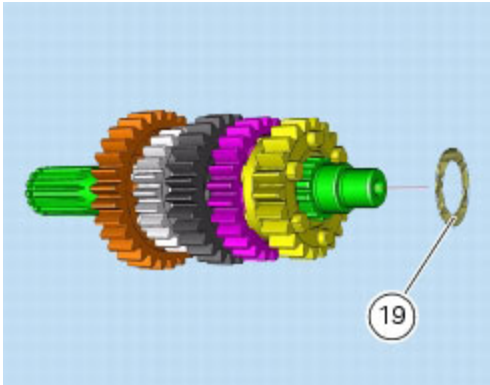
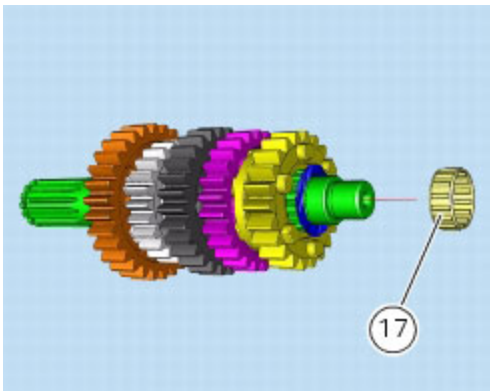
Disassembly of the gearbox secondary shaft

Remove the chain-side shim (15) and clutch-side shim (26) from the secondary shaft.

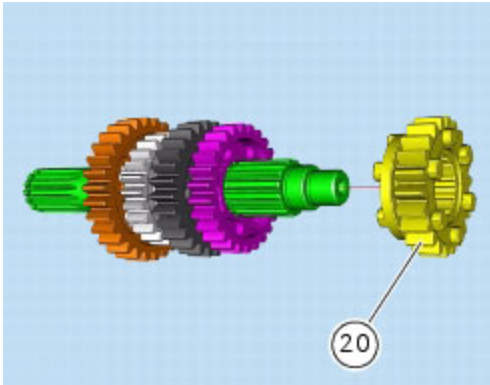


Withdraw the first speed driven gear (18) with the roller cage (17) and the shim (19).

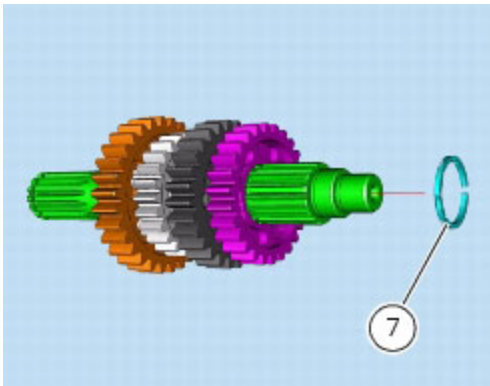


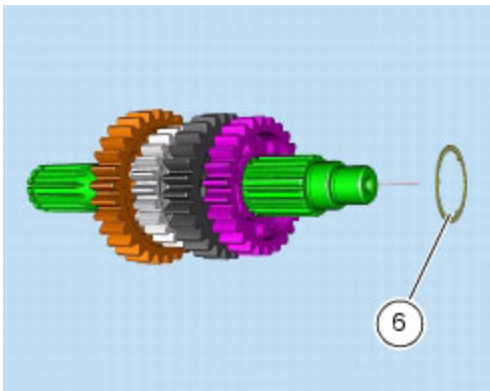


Remove the fifth speed driven gear (20).

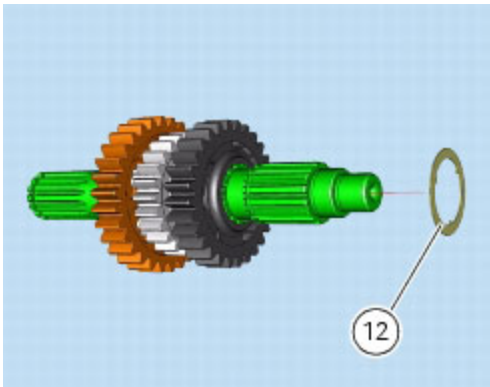
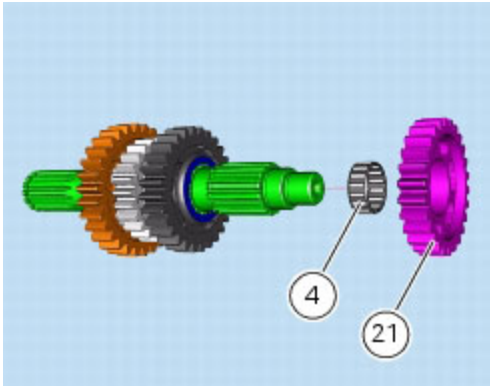


Use two flat blade screwdrivers to remove the circlip (7) taking care not to damage the shaft surface. Remove the circlip (7) and the splined washer (6).

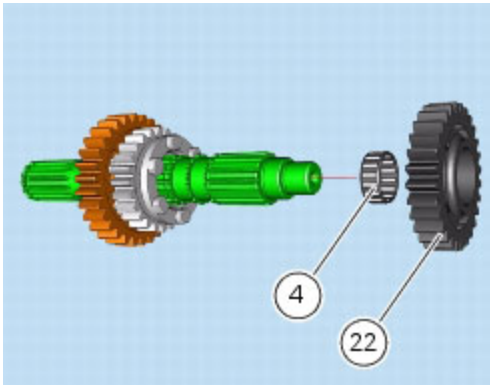


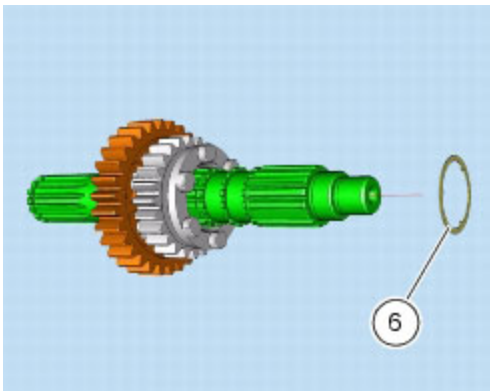


Withdraw the fourth speed driven gear (21) with the roller cage (4) and splined washer (12).

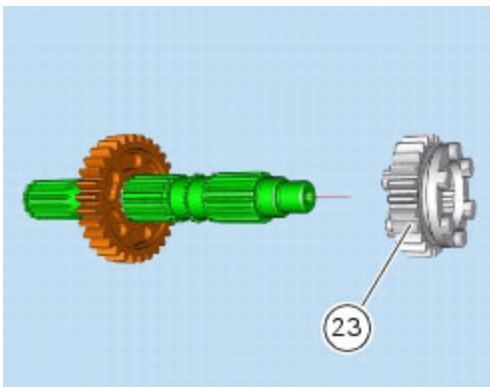
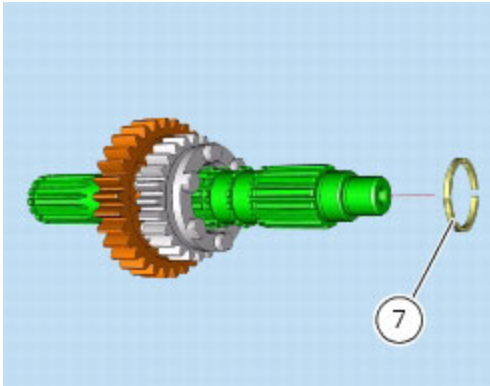


Remove the third speed driven gear (22) with the roller cage (4) and the splined washer (6).

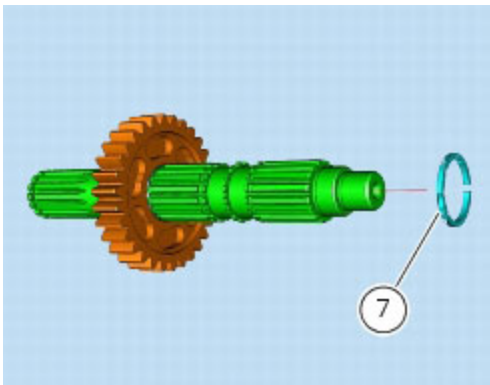


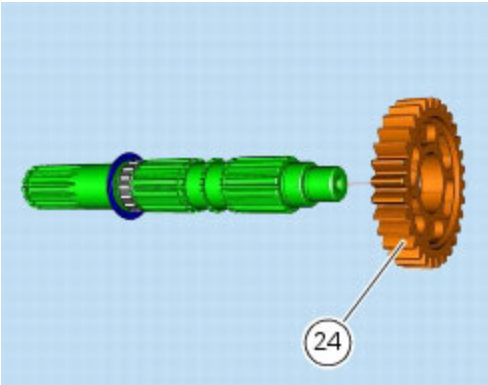
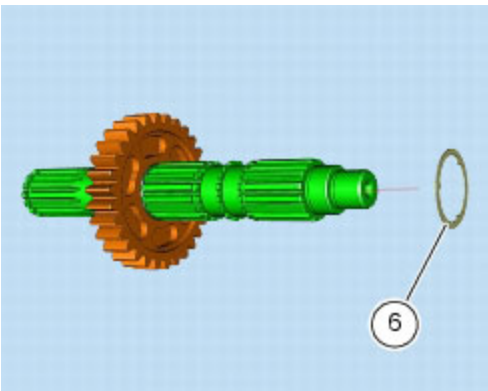


Remove the circlip (7) and remove the sixth speed driven gear (23).

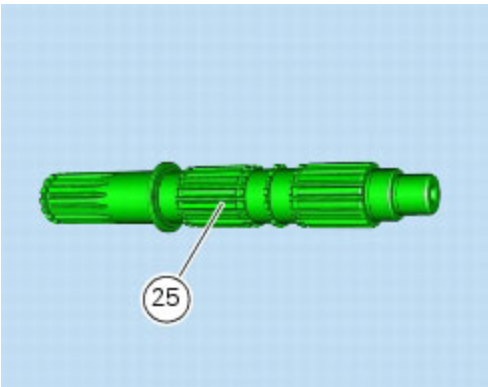
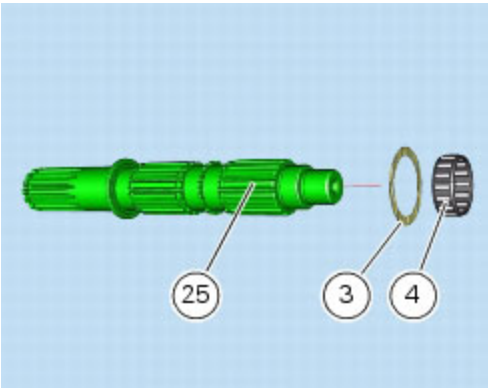


Remove the circlip (7) and withdraw the splined washer (6) and the second speed driven gear (24).

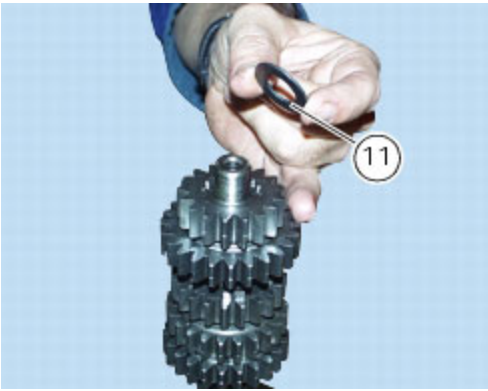




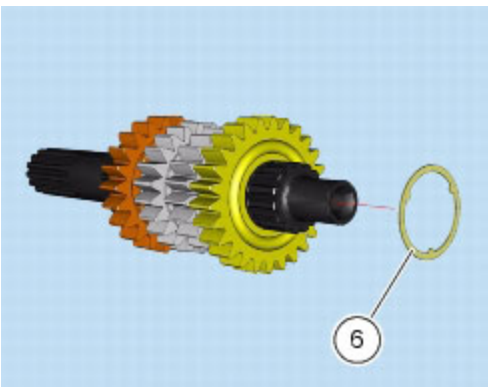
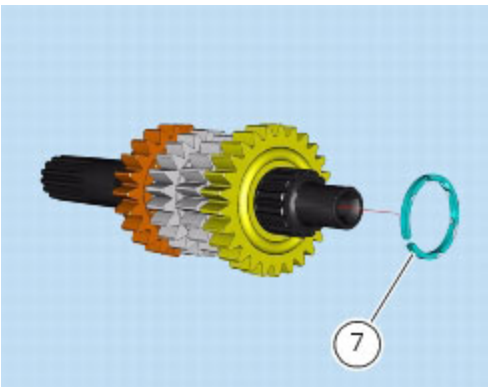
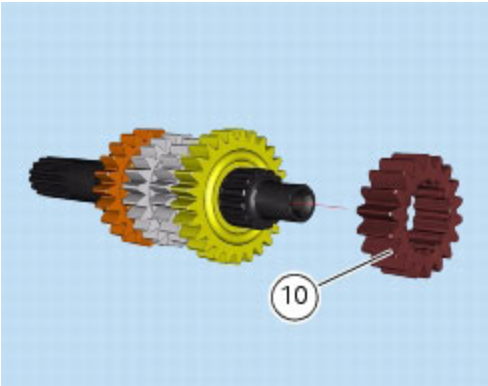
Withdraw the roller cage (4) and the shim (3). All the components have thus been removed from gearbox secondary shaft (25).



Disassembly of the gearbox primary shaft
 Remove the chain-side shim washer (11) and the clutch-side shim washer (**1**) from the primary shaft.



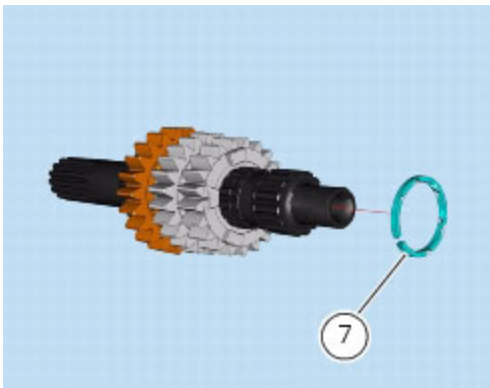
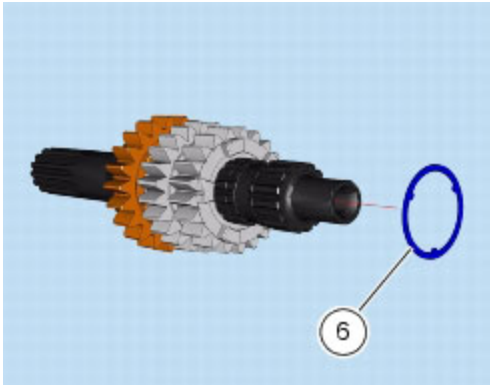
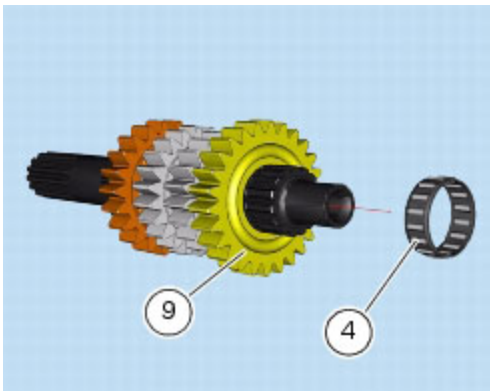
Remove the second speed driving gear (10). Use two screwdrivers to prise out the circlip (7) and the splined washer (6).



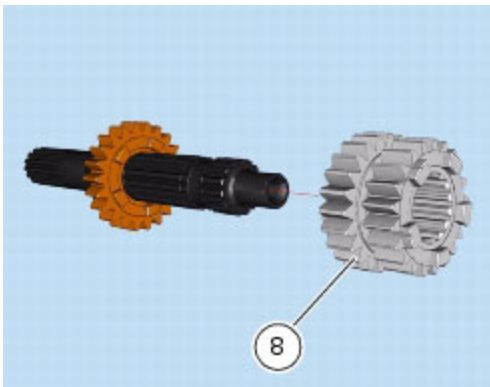
Important

Take care to avoid damaging the surface of the shaft while removing circlip (7).

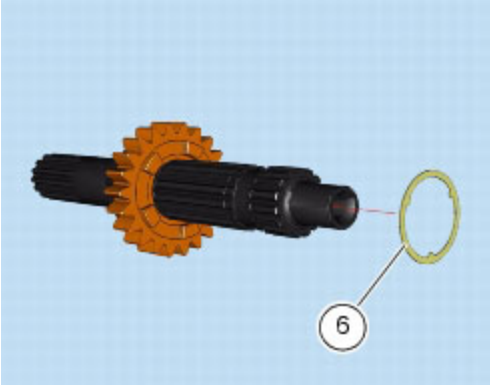
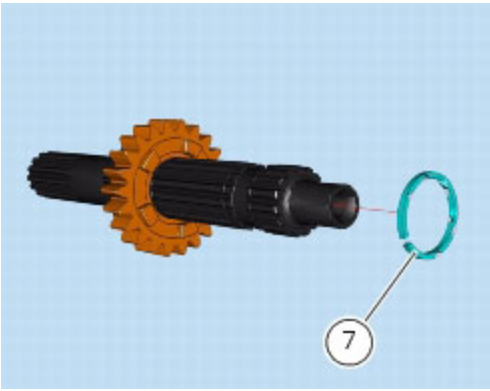
Remove the sixth speed driving gear (9) with its roller cage (4). Remove the splined washer (6) and the circlip (7).



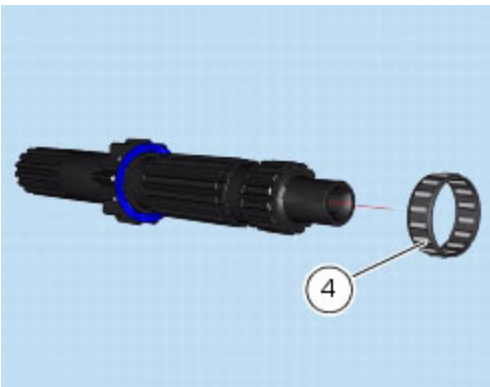
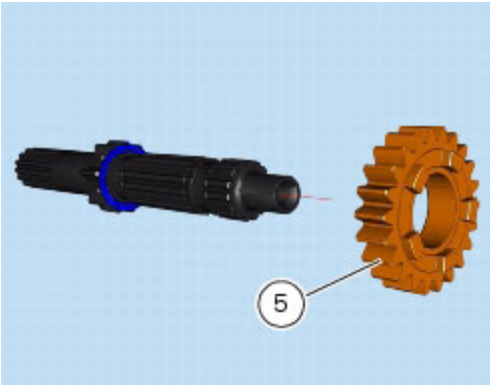
Withdraw the third and fourth speed driving gear (8).



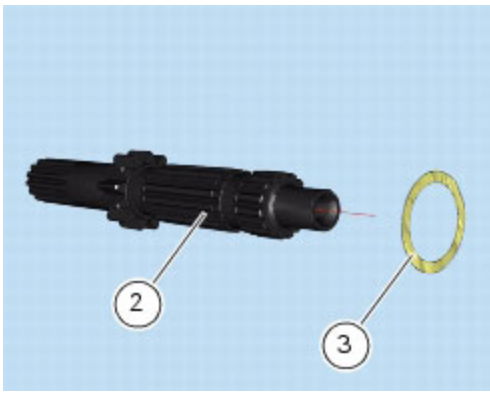
Remove the circlip (7) and the splined washer (6).



Remove the fifth speed driving gear (5) with the roller cage (4).



Slide the shim (3) off the primary shaft (2).



Overhaul of the gearbox

Check the condition of the front coupling dogs of the gears. They must be in perfect condition and with no sign of wear on the edges of the teeth.

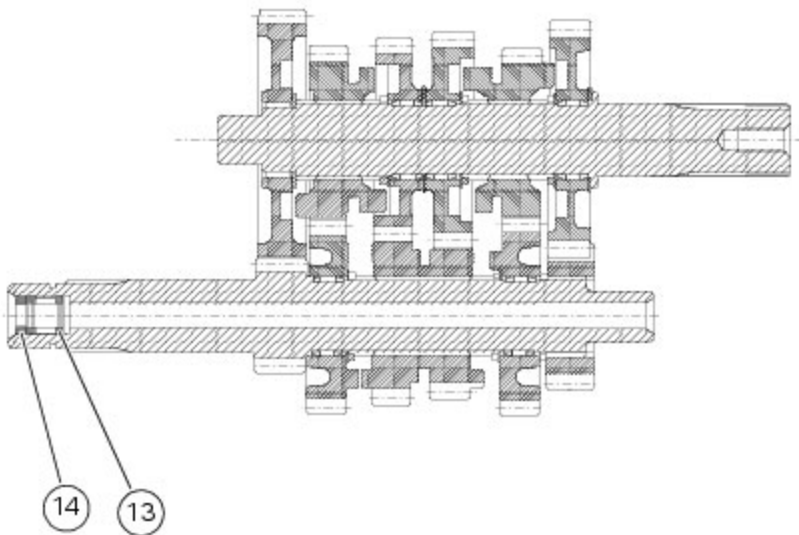
The idler gears must rotate freely on their shafts.

When refitting, make sure the circlips are correctly positioned.

Check the needle roller bearings for wear.

The threads and splines of the shafts must be in perfect condition.

At each overhaul, check the condition of the needle roller cage (13) and the oil seal (14) located on the end of the gearbox input shaft: remove them if necessary using a suitable puller and renew them.



Also check that the component parts of the gear selector mechanism are in good condition.

Engage the gears and check that the gearchange mechanism does not stick (selector fork - gear groove, and fork pin - desmodromic drum groove) due to incorrect end float. Restore the correct end float by shimming the gearbox shafts and the selector drum with suitable shims.

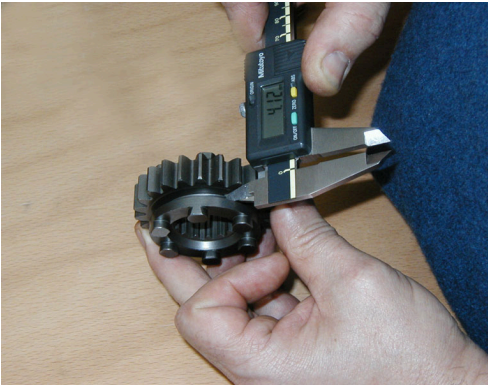
For the total gearbox shaft and selector drum end clearance values, refer to Sect. C 1.1 [Gearbox](#).

Inspection of the gear selector forks

Visually inspect the gear selector forks. Bent forks must be renewed as they may lead to difficulties in gear changing or may suddenly disengage when under load.

Use a feeler gauge to check the clearance of each fork in its gear groove.

If the service limit has been exceeded, check whether or not it is necessary to replace the gear or the fork by referring to the limits specified for each part (Sect. C 1.1 [Gearbox](#)).



Inspection of the gear selector drum

Use a gauge to measure the clearance between fork pin and the slot on the selector drum.

If the service limit is exceeded, determine which part must be replaced by comparing these dimensions with those of new components (Sect. C 1.1, [Gearbox](#)).

Also check the wear on the drum support pins; these must not show any signs of scoring, burrs, or deformation.

Turn the drum in the crankcase to establish the extent of radial play. If play is excessive, change whichever part is most worn.





Reassembly of the gearbox shafts

Figure 1 shows all the parts to be reassembled on the gearbox primary shaft (2), with the calculated end shims (1) and (11) (Sect. N 9.2, [Shimming the shafts](#)).

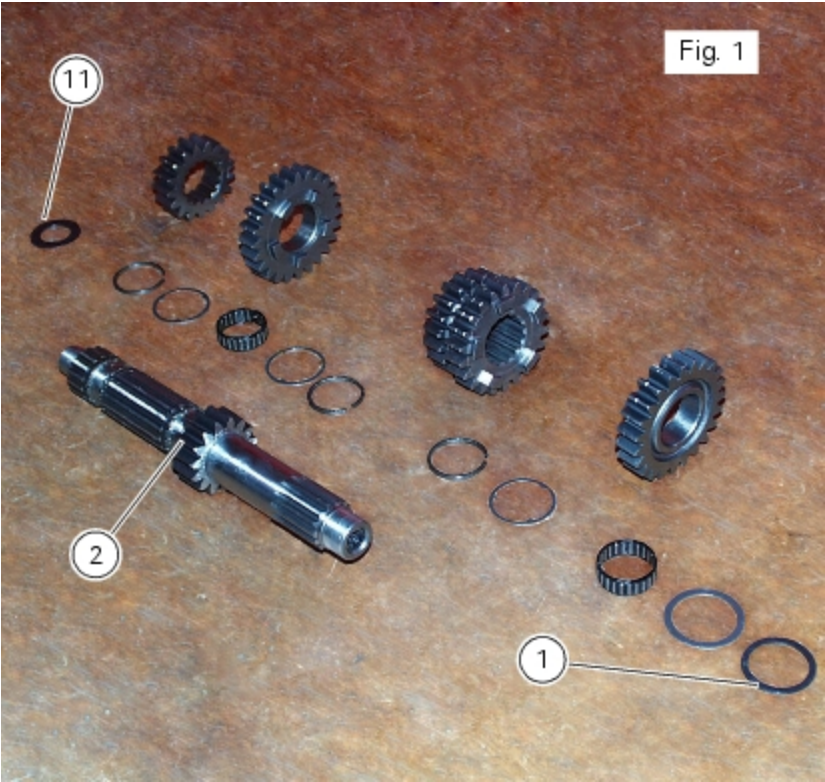


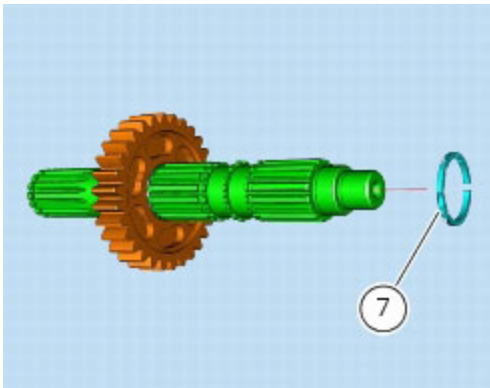
Figure 2 shows all the parts to be installed on the gearbox secondary shaft (25), with calculated end shims (15) and (26) (Sect. N 9.2, [Shimming the shafts](#)).



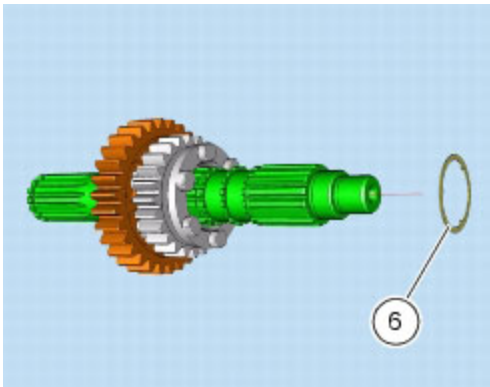
Reassemble the gears on the gearbox shafts by reversing the disassembly procedure.

Take particular care when installing the idler gears. The assembly of the 3rd and 4th speed gears and the relative fixing components on the secondary shaft is given as an example.

Fit the circlip (7), checking that it is fully inserted into its groove on the shaft. Push the circlip into position with a suitable size tubular drift.

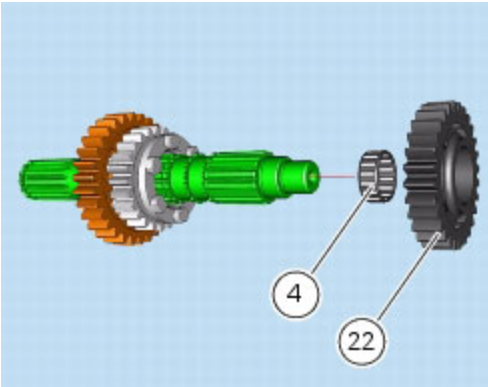
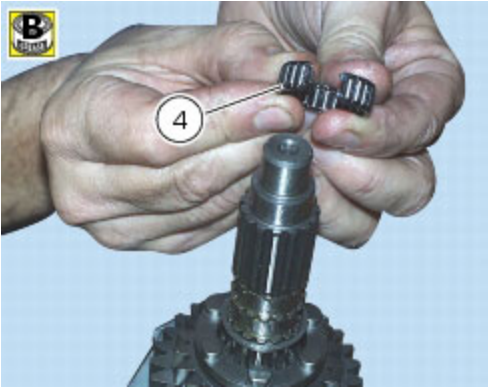


Slide the washer with three internal points (6) over the shaft until it locates against the circlip you have just fitted.

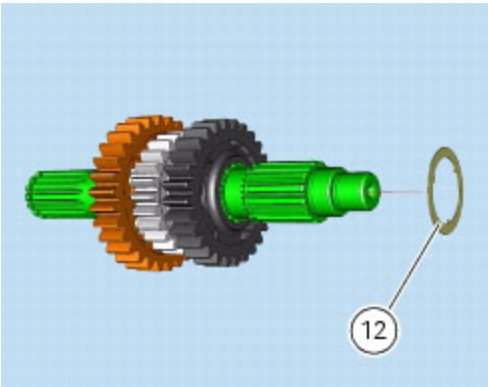


To fit the needle roller cage (4) onto the shaft, first lubricate it with plenty of grease (of recommended type) and then open it slightly to make it easier to slide on to the shaft.

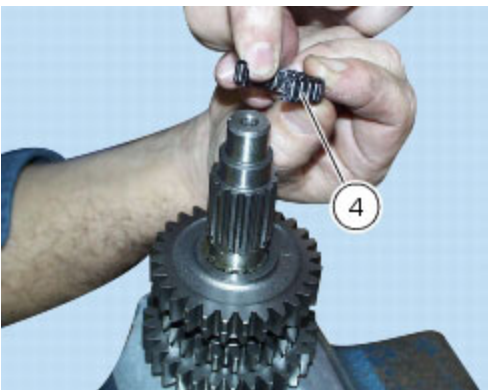
Fit the 3rd speed gear (22).

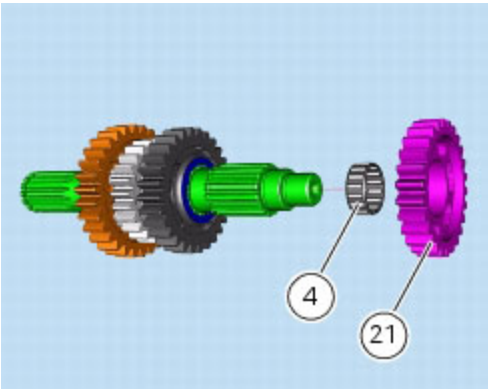


Fit on the gear the three-pointed washer (12), which can be distinguished from its counterpart (**6**) by its bigger outside diameter.

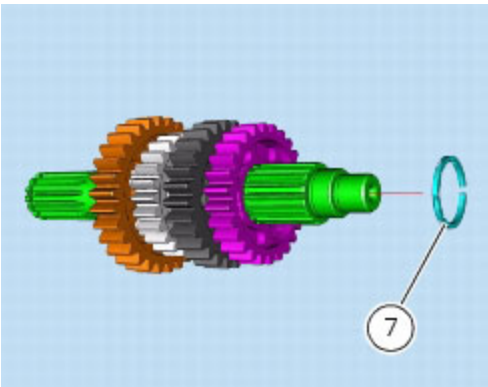
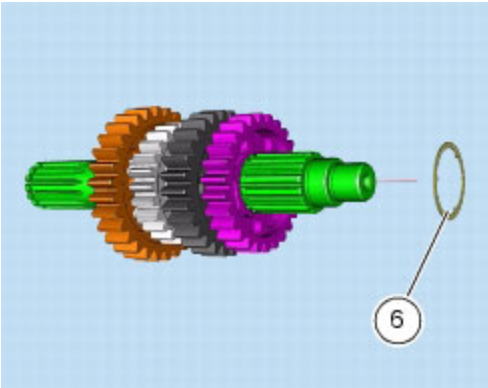


Fit another needle roller cage (4) using the method already described.
Fit the 4th speed gear (21).





Fit another safety washer (6) and another circlip (7) into the shaft. Seat the circlip using the drift used previously.

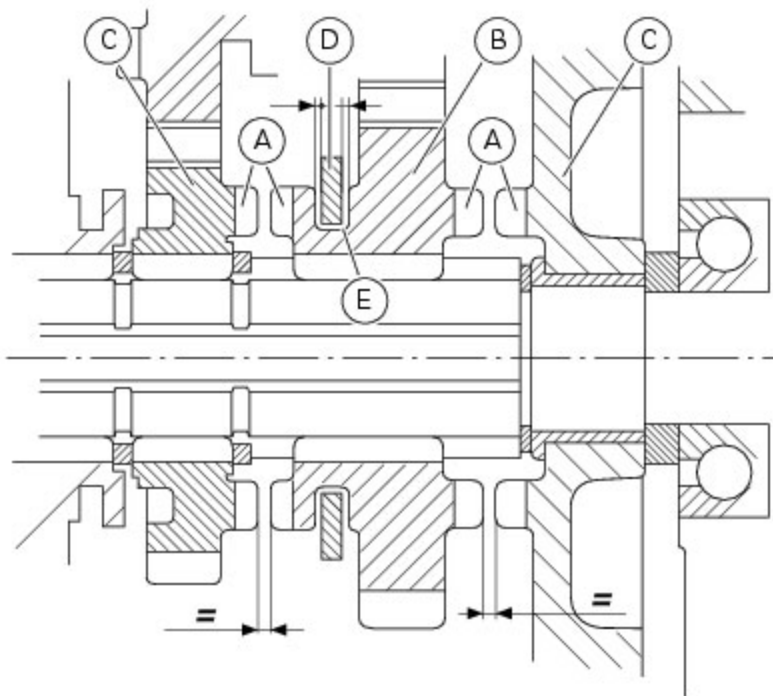


Reassembly of the gearbox

To refit the gearbox components follow the procedure under Sect. N 9.2, "[Reassembly of the crankcase halves](#)", relating to reassembly of the engine crankcase.

As a final practical test, ensure that with the gearbox in neutral the front coupling dogs (A) of sliding gears (B) are equidistant on both sides with respect to the corresponding coupling dogs on the fixed gears (C).

Check also that there is always a small amount of clearance between fork (D) and relative groove (E) on sliding gear (B) when engaging the gears.



Operation	Section reference
Reassemble the engine crankcase halves	N 9.2, Reassembly of the crankcase halves
Refit the starter motor	P 3, Refitting the starter motor
Refit the starter motor idler gear	N 9.1, Refitting the starter motor gear
Refit the primary gear	N 6.3, Fitting the primary drive gear pair and checking backlash
Refit the clutch-side crankcase cover	N 6.2, Refitting the clutch cover
Refit the clutch assembly	N 6.1, Reassembling the clutch
Refit the mesh filter	D 4, Changing the engine oil and filter cartridge
Refit the alternator-side crankcase cover and alternator assembly	N 8, Refitting the generator cover
Refit the cylinder barrel/piston assemblies	N 5, Refitting the cylinder/piston assembly
Refit the complete cylinder head unit	N 4.4, Refitting the cylinder head assembly
Refit the camshafts	N 4.3, Refitting the camshafts
Refit the timing belts and the timing belt pulleys	N 4.2, Refitting the timing belt covers
Refit the lubrication system	N 2.1, Refitting the oil pump
Refit the engine to the frame	N 1, Refitting the engine